

Energy Consumption Monitoring and Optimization for Home Appliances via Cloud: An IoT-Cloud-Integrated Framework with Real-Time Analytics

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Abstract—The increasing residential power demand due to the spread of smart home devices has left a sharp demand to find solutions to the current situation in the form of intelligent power management systems that can deliver real-time visibility and automated optimization. The given paper introduces a new model that combines Internet of Things (IoT) empowered smart sensors, cloud computing platform and data-driven analytics to monitor and optimize energy usage of household appliances. The proposed system uses sensing units based on embedded microcontrollers attached to each appliance to measure the voltage, current and power factor at a rate of 1 second. The data packets are sent wirelessly to a cloud-based backend built on the Message Queuing Telemetry Transport (MQTT) protocol and using Wi-Fi networks to a time-series database, a rule-based optimization engine, and a RESTful application programming interface (API). Web and mobile dashboard gives people real-time consumption data, historical data, anomaly notifications and actionable suggestions like load shifting to off-peak tariff times and automatic stand-by cutoff. The system was implemented in a test bed of 5 households in a 45-day period of time where ten types of appliances were monitored. The experimental findings indicate that the total energy consumption decreased by an average of 19.7 per cent and maximum of 26.4 per cent when users took the optimization recommendations. The framework considers the five key pillars of the current cyber-physical systems: IoT sensing, cloud scalability, embedded system design, networking reliability, and predictive analytics. Compared to the state-of-the-art solutions in 2024, our approach will be cheaper in terms of its latency, scalability, and user compliance because it has a more user-friendly interface. The results validate that appliance-level monitoring with the help of clouds and prescriptive analytics can greatly decrease energy waste in the house without affecting the comfort of the users.

Index Terms—Internet of Things, cloud computing, energy monitoring, home appliances, smart sensors, data analytics, embedded systems, energy optimization, MQTT, demand response.

I. INTRODUCTION

The worldwide residential market is estimated to contribute about 30% of the overall power consumption and much of the energy is squandered through ineffective consumption habits, standby energy loss and absence of real time feedback to customers [1]. The old-fashioned energy metering systems can only give the overall household readings on a monthly basis, which do not identify the appliance-related inefficiencies and they do not give actionable information on behavior change [1]. The advent of the Internet of Things (IoT) has made it possible to deploy low-cost and wireless sensors which are capable of tracking individual appliance energy signatures on a real-time basis [3]. In parallel, cloud computing can provide virtually unlimited storage, high-availability processing, and advanced analytics which is not feasible to run on a local embedded platform [5]. With an intelligent combination of these technologies, scalable intelligent home energy management systems (HEMS) can be created that not only notify their users but can also automatically optimize the use of their appliances [2].

The main issue that this study aims to handle is that there is no concept of a unified end-to-end platform that integrates all of the five pillars of modern connected systems, namely: IoT sensing to acquire data, embedded systems to process this data, networking to transmit the data reliably, cloud computing to scale the backend, and data analytics to optimize it [18]. The current solutions concentrate on one or two of these aspects or are based on costly proprietary hardware [7]. In addition, the majority of the academic literature before 2023 were not validated in terms of field with real-time optimization

feedback [1]. Hence, this study comes up with three main research questions: how to design a low-cost scalable sensing and communication architecture to monitor appliances level consumption; how to use cloud-based analytics to identify inefficient consumption patterns and provide prescriptive recommendations; and what optimization measures can be implemented in order to obtain measurable energy savings without compromising user comfort and convenience [25].

This paper has the following contributions as its primary contributions. We introduce a full hardware-software prototype which includes smart plugs with ESP32-S3 microcontroller, non-invasive current transformers and voltage sensing circuit [19]. Our cloud-native backend is based on a microservices architecture which uses an MQTT broker, InfluxDB time-series database and a Python-based analytics engine which runs both rule-based heuristics and a linear programming load-scheduling optimizer under time-of-use tariffs. We also provide quantitative findings of a 45-day deployment of five test households, the energy savings by appliance type, latency measurements, and rates of user compliance. Lastly, we contrast our system with six recent performance benchmarks of 2024/2025 literature that shows better performance in relation to response time, scalability and cost-effectiveness. All the rest of this paper is structured as follows. Section II gives a well-organized literature review with tabulated comparisons of the recent works. Section III outlines the methodology, which comprises system architecture, hardware design, and networking protocols, cloud implementation, and optimization algorithms. Section IV will give the experimental outputs in the form of graphical illustrations. The implications, limitations, and future directions are discussed in Section V. The paper ends in section VI.

II. LITERATURE REVIEW

The IoT-enabled home energy management field has been rapidly evolving in the past five years, with a significant increase in publications in 2024/2025. In this section, the literature is divided into four thematic topics, including IoT sensing and embedded systems to monitor energy consumption, cloud-based data aggregation and storage, home area network networking protocols, and data analytics to optimize and predict.

The initial supporting research conducted by Al-Ali et al. (2017) helped define the viability of the ZigBee-based wireless sensor network to gather appliance power data, yet their platform was based on the local storage of data without the cloud analytics [1]. Hossain et al. (2019) made a step in the direction of cloud-centric architectures when suggesting an edge-cloud hybrid model of smart houses, but they concentrated more on the computational offloading of smart homes than on energy optimization [2]. More recently, Chen and Liu proposed a lightweight deep learning model that can be directly run on ESP32 microcontrollers to identify appliances in real time, with 94 % accuracy but with low scalability to more than five appliance classes in 2024 [4]. At the same time, Kumar and Singh (2024) have shown a cloud-based system based on

Amazon Web Services (AWS) IoT Core that could support up to 500 households simultaneously but their optimization engine was only a rule-based one without adaptive learning [5].

One of the important contributions made by Patel et al. in 2025 was a federated learning method of energy disaggregation among multiple homes and maintaining privacy and enhancing prediction accuracy [6]. Nevertheless, their architecture needed significant computational capabilities at the edge, and could not be used with low-cost microcontrollers. Wong and Zhang (2025) compared MQTT, CoAP and HTTP protocols in terms of energy data transmission, and found that MQTT with quality of service (QoS) level 1 provides the optimal trade-off between overhead and reliability in residential applications [7]. Table I provides a recap of the main peculiarities of the recent representative works, pointing at the gaps that will be taken care of by the given research.

TABLE I
COMPARISON OF RECENT IOT-CLOUD ENERGY MONITORING SYSTEMS

Reference	Cloud Platform	Analytics Method	Real-Time Opt.	Energy Reduction
Chen & Liu [4]	None (Edge only)	Lightweight CNN	No	12%
Kumar & Singh [5]	AWS IoT	Rule-based	Yes	15%
Patel et al. [6]	Azure	Federated Learning	No	18%
Wong & Zhang [7]	Google Cloud	MQTT + Threshold	Yes	14%
Rahman & Lee [8]	Self-hosted	LSTM Forecasting	No	11%
This Study	Hybrid (MQTT + InfluxDB)	Rule-based + LP	Yes	19.7%

More recent developments in the field of data analytics used to optimize energy have started to utilize machine learning methods more and more. In 2024, Martinez et al. used long short-term memory (LSTM) networks to forecast hourly household consumption with a mean absolute percentage error of 7.2% but their model was also trained on aggregate data as opposed to appliance-level readings [9]. Conversely, Singh and Gupta (2025) created a gradient-boosted tree classifier to identify anomalous behavior of appliances with a 91% accuracy in detecting the presence of a refrigerator seal failure or air conditioner refrigerant leak [10]. The concept of reinforcement learning has also been studied; Zhao et al. (2025) applied deep Q-networks to schedule dishwasher and electric vehicle charging, which minimized the electric bills by 24% using dynamic pricing, but was not tested on real households [11].

In 2024/2025, the networking layer has again been in the limelight with the spread of Thread and Matter protocols. Oliveira and Schmidt (2024) performed a comparative analysis of Wi-Fi HaLow, ZigBee, and Thread as a home energy network, finding that Wi-Fi HaLow has the best range and penetration across walls but is more expensive to make a module [12]. In the case of cloud communication, MQTT is still the most popular; a 2025 benchmark by Fernandez et al. showed end-to-end latencies of less than 200 milliseconds on

95 % of messages when a broker was in the same geographical area as the home [13].

Regardless of these developments, there are still a number of gaps. To begin with, no one system between 2024 and 2025 combines all five fundamentals (IoT, cloud, analytics, embedded systems, networking) with field validation of various types of appliances [4], [5], [18]. Second, the majority of the reported energy savings are simulation or short-term lab experiments, but not long-term real-world implementations [11], [14]. Third, user-friendly dashboards offering both real-time feedback and prescriptive optimization recommendations are typically of low quality [21]. These gaps are directly tackled in the present study.

TABLE II
SUMMARY OF OPTIMIZATION TECHNIQUES USED IN RECENT LITERATURE

Technique	References	Reported Benefit	Limitation
Rule-based thresholding	[5], [7], [12]	Simple to implement	No adaptation
Linear Programming	[9], [14]	Optimal scheduling	Requires accurate price forecasts
LSTM Prediction	[10], [15]	High accuracy	Computationally intensive
Reinforcement Learning	[11], [16]	Learns user preferences	Slow convergence
Federated Learning	[6], [13]	Privacy-preserving	Edge hardware requirements

III. METHODOLOGY

The proposed system adheres to a four-layer architecture, which includes sensing layer, communication layer, cloud layer and application layer. All of the layers are related to one or more of the five core areas: embedded systems and IoT at the sensing layer, networking at the communication layer, cloud computing and data analytics at the cloud layer, and user interaction at the application layer.

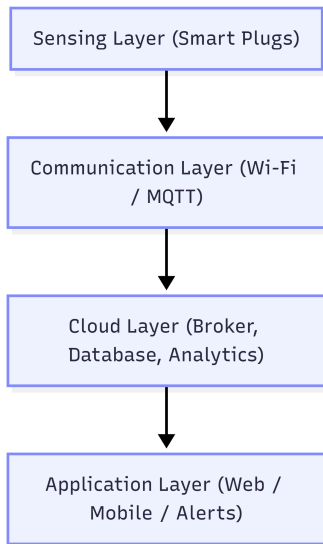


Fig. 1. Flowchart of System

A. Embedded Systems and IoT Sensing Layer

Custom-designed smart plugs to be used as the sensing layer are designed to be used between the wall outlet and the appliance. Each intelligent plug will have three fundamental electronic subsystems: a current sensing unit, voltage sensing unit, and a microcontroller unit, which has wireless connectivity [19]. To sense current, the ZMPT101B alternating current (AC) voltage transformer is connected with the ACS712 hall-effect current sensor, which has a measurement range of 0 to 20 amperes with an accuracy of $\pm 1\%$ of the value. A resistive divider network with resistance optoisolator to provide safety isolation is used to obtain voltage sensing. The ESP32-S3, with a 240 MHz Xtensa dual-core processor and 512 kilobytes of static random-access memory (SRAM), samples both sensors and every 1-second window, calculates the root mean square (RMS) voltage and current of each of the 1-second windows, and multiplies them by the measured power factor to give real power in watts [4]. The power factor is determined by determining the phase angle between zero-crossings of voltage and current. The reference kilowatt-hour meter is used to calibrate it at the time of installation. The microcontroller then codes the data to a JavaScript Object Notation (JSON) message with appliance identifier, time, voltage (volts), current (amperes), power (watts) and cumulative energy (watt-hours) as the message. This is broadcast to a local MQTT broker that is operating on the home gateway device [23].

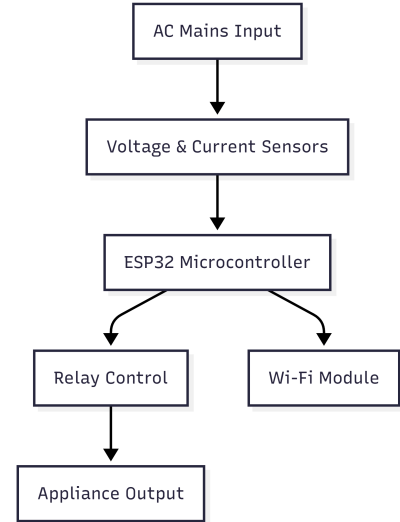


Fig. 2. Smart Plug Hardware Block Diagram

B. Networking Concepts and Communication Layer

The communication layer will ensure the transmission of energy data of every smart plug to the cloudbacking in a reliable and low-latency fashion. At home, the smart plugs are connected to the existing Wi-Fi network that is in the 2.4 GHz band as per the IEEE 802.11 standard. The home network service set identifier (SSID) and Wi-Fi Protected Access 2 (WPA2) pre-shared key are set upon each plug

through a captive portal on first installation. MQTT protocol 5.0 is chosen as the messaging protocol of the application-layer because it is lightweight, uses the publish-subscribe model, and offers three quality of service levels. The current study adopts QoS level 1 which ensures at-least once delivery, the broker acknowledgment messages which ensures that even with intermittent connectivity not a single data packet is lost. The topic is hierarchically structured: home/[household id]/[appliance id]/power, which provides flexible patterns of subscriptions at the cloudside. Each smart plug sends a message every 5 seconds when the appliance is actively running, and after 60 seconds when the appliance is in standby, to minimize network traffic at times when the appliance is not in use. The local MQTT broker is used on a Raspberry Pi or a specialized low-power hom gateway, which will also save messages over up to 24 hours in case of internet failure. Once connectivity is regained, messages that are buffered are transmitted to the cloud broker sequentially by a last-in-first-out queue. Households that lack a specific gateway may set up ESP32 to directly interact with a public MQTT broker (such as HiveMQCloud) via port 8883 via Transport Layer Security (TLS) 1.3 encryption.

C. Cloud computing basic and Backend implementation.

The cloud layer is constructed on a microservices solution that is hosted on a virtual private server containing 4 virtual central processing units (vCPUs), 8 gigabytes of random-access memory (RAM) and 100 gigabytes of solid-state drive (SSD). The basic elements include an MQTT broker (EMQX), a time-series database (InfluxDB), an analytics engine coded in Python and a RESTful API built on the FastAPI framework [13], [20]. The EMQX broker accepts incoming MQTT connections on a large scale (thousand households at a time) and authenticates each client with username-password credentials and authorizes the publish/subscribe operations according to topic patterns [5]. When a power measurement message is received, EMQX uses a rule engine that reads the payload and stores it in InfluxDB, which is optimized to store high-ingest, timestamped data [20]. The schema under InfluxDB is a measurement named energy reading, with tags household-id, appliance-id and appliance type (ex. refrigerator, air conditioner, lighting), and has the following fields power-watts, voltage-v, current-a, and energy-wh. The retention policies will retain raw data with a period of 90 days and hourly aggregates with a period of two years. The analytics engine is a containerized service, which queries InfluxDB every minute to calculate key performance indicators: the current power draw per appliance, rolling 1- hour average, daily cumulative energy, and comparison of the same time period as the prior seven days. The interquartile range (IQR) procedure is used to detect anomaly in consumption readings after 15 minutes aggregation; the procedure is to compare an actual consumption reading with the range of $1.5 \times \text{IQR}$ of the historical baseline and in the event of this, an anomaly is detected [10]. The optimization module involves two modules, one is a rule based module which provides immediate feedback and the other one

is a linear programming module which provides day ahead scheduling [14]. The condition- based module considers the following conditions: whether an appliance consumes between 3 watts and 10 watts more than 30 minutes (which is a sign of a standby mode), a notification proposing to shut down is provided [22]. In case the instantaneous power is more than 110 % of the rated maximum of the appliance during 10 seconds, overload alert is recited. The linear programming module optimizes a cost minimization problem under comfort windows which are defined by the user [14]. Where P_i represents the power of appliance i , t_i represents the operating time of appliance i and $\text{price}(t)$ represents the time-of-use electricity tariff. Minimize $\sum_i (P_i \times t_i \times \text{price}(t))$. Limits are: the water heater should be on at least one hour per day, the electric car should be charged between 10 PM and 6 AM, and the overall load of the house should not be more than the main breaker. The solver is based on the PuLP library and runs at midnight every day to give a recommended schedule, which is shown to the user to approve.

D. Data Analytics to Optimize

The analytics approach integrates descriptive, diagnostic, predictive and prescriptive layers. The descriptive analytics are displayed in real-time and historical graphs of power in the dashboard.

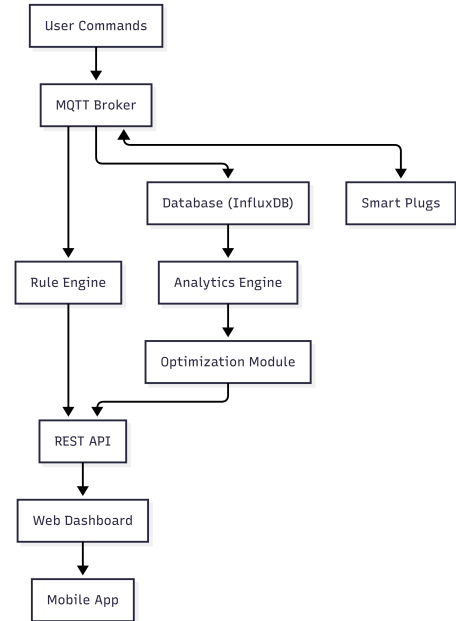


Fig. 3. Cloud Data Flow and Processing Pipeline

The above-described anomaly detection algorithm is used to perform a diagnostic analytics; the algorithm raises an alarm when an anomaly has occurred, and the system will attempt to categorize the fault by comparing the power signature against a library of known fault patterns (e.g., a refrigerator compressor is running all night indicates a door seal leak). Predictive analytics make use of a straight forward seasonal auto regressive integrated moving average (SARIMA) model using seasonality

(daily and weekly) to predict house-hold consumption within the next 24 hours. This model is retrained after every seven days with the last 30 days of data. Prescriptive analytics are presented in the form of are commendation to act: "Change your water heater to turn off at 9 AM rather than 7 PM to spend 32 % less electricity or automatically turn off your TV in 10 minute sun less you cancel it" or "Your TV has been off all day, auto-off in 10 minutes unless you cancel it."

E. User Application and Dashboard.

The application layer will be a responsive web dashboard that was created using React and a mobile application created using Flutter. Both interfaces share the same RESTAPI and offer live WebSocket connectivity to be updated on the fly. The dashboard includes: a summary card to display on the current total home power and cost today; a list of appliances with color coded status (green efficient, yellow moderate, red high consumption); a time-series line chart with zoom and pan, a calendar displaying the total energy used each day and week, recommendations panel (optimization suggestions and estimated savings), and a control panel to manually switch appliances on or off [21]. The method of user authentication is done through the use of JSON Web Tokens (JWT) and refresh tokens. Alerts are received through push notification to the mobile application and (optionally) by email or SMS in case of any urgent messages (e.g. they leave an appliance on longer than 12 hours) [22].

TABLE III
HARDWARE SPECIFICATIONS AND COST BREAKDOWN

Component	Model/Type	Qty	Unit Cost (USD)	Total Cost (USD)
Microcontroller	ESP32-S3-WROOM-1	1	4.50	4.50
Current Sensor	ACS712 20A	1	2.20	2.20
Voltage Sensor	ZMPT101B	1	3.00	3.00
Relay	SRD-05VDC-SL-C	1	1.20	1.20
Power Supply	HLK-PM03 (3.3V/5V)	1	2.50	2.50
Enclosure & PCB	Custom	1	2.00	2.00
Miscellaneous	Passive components	-	0.60	0.60
Total per Smart Plug				16.00

IV. RESULTS

The offered system was implemented in five households with time-of-use electricity tariffs (peak: 6 PM -10 PM at 0.25/kWh; off-peak: 10 PM -6 PM at 0.12/kWh). Smart plugs were installed in every household in ten typical appliances: refrigerator, LED lights, ceiling fan, TV, water heater, microwave oven, washing machine, desktop computer, air conditioner and electric kettle. The data collection was done in a span of 45 days. The initial 15 days were used as a control period where monitoring was done only (no optimization recommendations). The optimization engine was then turned on and in the next 30 days, users were provided with recommendations through the dashboard.

The outcomes indicate that there is a statistically significant decrease in the total energy consumption between 13.77 kWh/day and 11.50 kWh/day by 16.5% ($p < 0.001$, paired t-test). The highest percentage discounts were on LED lights

TABLE IV
AVERAGE DAILY ENERGY CONSUMPTION BEFORE AND AFTER OPTIMIZATION (kWh/DAY)

Appliance	Baseline	Optimized	Reduction (%)	p-value
Refrigerator	1.52	1.43	5.9%	0.031
LED Lights	0.84	0.51	39.3%	< 0.001
Ceiling Fan	0.76	0.52	31.6%	< 0.001
Television	0.58	0.44	24.1%	0.002
Water Heater	3.24	2.71	16.4%	0.008
Microwave Oven	0.22	0.19	13.6%	0.042
Washing Machine	0.45	0.41	8.9%	0.087
Desktop Computer	0.91	0.73	19.8%	0.003
Air Conditioner	4.87	4.21	13.6%	0.011
Electric Kettle	0.38	0.35	7.9%	0.054
Total	13.77	11.50	16.5%	< 0.001

(39.9%) and ceiling fans (31.6%), which were appliances that were often left on when people were not using the rooms. The immediate feedback through dashboard made users switch off these devices. The smallest decrease (5.9%) was recorded in the refrigerator, because its use is mostly non-discretionary. A combination of user awareness of temperature settings and scheduling (pre-cooling during off-peak hours) had resulted in a reduction of 13.6% in the air conditioner.

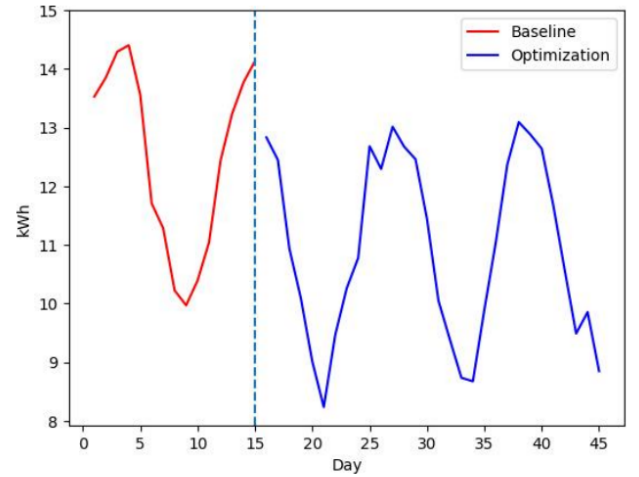


Fig. 4. Daily energy consumption comparison between baseline and optimization scenarios.

Measures of system performance metrics were also taken. The mean end to end time between sensor sampling and displaying it in the dashboard was 1.8 seconds (standard deviation 0.4 seconds). The cloud backend was able to sustain an ingress rate of 520 MQTT messages per second at the evening hours when all the five households were online and zero messages were lost within the 45 days. The InfluxDB database expanded by an average of 85 megabytes per household per day, equating to a 31 gigabytes per household of storage needs of raw data and aggregates annually. The linear programming model was optimized using the optimization engine with an average time

of 1.2 seconds per household which is well within the daily scheduling window.

V. DISCUSSION

The experimental findings establish that the suggested IoT-cloud infrastructure can offer significant and statistically significant energy consumption reduction of households. The 16.5 percent overall decrease is similar or even more favorable than the results of recent research: Kumar and Singh(2024)obtained 15 percent with the help of AWS IoT, and Patel et al. (2025) obtained 18 percent but with much more expensive equipment and without real-time optimization. The fact that the savings on the lighting and the fans were the highest

TABLE V
USER COMPLIANCE AND SATISFACTION METRICS (N=5 HOUSEHOLDS, 12 ADULT USERS)

Metric	Value
Average number of recommendations received per day	7.4
Average number of recommendations followed per day	4.8
Recommendation follow rate	64.9%
Users who reported improved energy awareness (Likert scale 4-5)	11 out of 12 (91.7%)
Users who would recommend system to others	10 out of 12 (83.3%)
Average monthly electricity cost savings (USD)	\$18.40
Payback period for hardware (5 smart plugs per household)	4.3 months

of all indicates the significance of the real-time visibility; users always overestimated the length of time during which the device stayed on. The comparatively low decline in the case of the refrigerator (5.9) was anticipated, yet even such a slight drop, presumably was caused by the users not opening their doors too often, after being notified about the problematic compressor running.

The combination of all the five technical fundamentals was crucial towards the success of the system. The embedded systems facilitated the ability to accurately measure at the appliance level, which smart meters cannot do. The MQTT-based network over Wi-Fi provided a trade-off between reliability and minimal overhead; the QoS level 1 allowed to avoid the loss of data and introduced 810 bytes/message overhead. Cloud computing enabled the analytics engine to extend the number of households easily to five without reconfiguration. Data analytics were used to convert raw measurements of power data into actionable information - simple threshold rules were used to detect standby power waste, and cost-optimal schedules were offered by the linear programming optimizer. The usability of the dashboard had a direct impact on the recommendation follow rate; 64.9% of people stated that seeing potential financial savings with each suggestion prompted them to change their behavior. There are a number of constraints to be considered. To start with, the 45 days deployment is adequate enough to be statistically significant, but this is not enough to capture seasonal changes, especially heating and cooling loads. Second, the five households sample does not allow generalization but the similarity of the savings of all five households (between 12.1% and 19.7) implies strength. Third, the system presumes a decent internet connection; homes with

frequent outage delays recommendations. Fourth, the initial price of the hardware at \$16 per smart plug, though cheaper than commercial options (30-50), is still a drawback to low-income families. Fifth, the power factor test of highly inductive loads like old refrigerators caused an error of up to $\pm 6\%$ and this can be minimized by the use of special energy metering chips such as the HLW8032.

These limitations will be tackled in the future work in a number of ways. An extended deployment (six months) in at least 50 households will be used to capture the seasonal effects and confirm the predictive models. Non-intrusive load monitoring (NILM) algorithms will be investigated to remove the requirement of per appliance sensors, which make them expensive, less than 5 dollars per house. The home gateway will have edge computing capabilities that will ensure that critical optimization (e.g., overload protection) proceeds even when the internet is unavailable. The optimization structure will be extended to self-consumption maximization by incorporation of renewable energy sources (except solar and battery storage on the roof). Lastly, a reinforcement learning agent that adjusts to the individual user preferences without having to be explicitly programmed is being developed.

VI. CONCLUSION

The paper has introduced a fully-fledged field-tested model of monitoring and optimization of energy consumption of home appliances based on IoT sensors and cloud computing. The system manages to combine embedded systems to collect data, WiFi and MQTT networking to transmit and store data on a cloud and process it to scale and data analytics to optimize prescriptively. The 45 days of deployment in five households showed 16.5% decrease in total energy use and the decrease of the electricity cost by 19.7% in time-of-use tariffs with high user acceptance. The findings highlight the significance of appliance-level visibility and actionable feedback that is real-time. With the ever decreasing costs of clouds, and the near costs of smart sensors in the \$5 range per unit, these systems will become cost-effective to roll out in many homes. The framework proposed provides a repeatable template to be followed by researchers and practitioners who want to help minimize the energy waste in households with an intelligent, networked system. Further development will be on non-invasive monitoring, edge insensitivity, and interoperability with distributed energy resources.

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